

Rishikesh S

Founding Software Engineer | Full-Stack Systems, Applied AI & Cloud Infrastructure

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Summary

Full-stack software engineer with **4+ years** shipping production systems across backend, frontend and infrastructure. Founding engineer at Mando, an AI platform for Workday customers: took **five product surfaces** from first commit to the company's **first paid tier** in under two years as sole engineer. Previously built real-time trading systems at Quantitative Brokers under strict latency and reliability budgets.

Work Experience

Founding Software Engineer

Sep 2024 – Present

Mando

Remote

- Engineered **conversational search** over Workday's knowledge base, owning retrieval, ranking and **cited answer generation (RAG)**, so users get a verifiable answer, not a list of documents.
- Shipped the **expert consultation product** end to end: time-zone-aware booking, live video and recording on Azure Communication Services, and a browser-based editor for trimming and annotating footage.
- Built the **service-management platform** behind it (ticket intake, routing, expert assignment, resolution) and a **Chrome extension** that records a workflow once and generates step-by-step documentation from it.
- Launched Mando's **first paid tier** by integrating Stripe subscriptions, checkout, invoicing and entitlements.
- Architected the full stack as sole engineer: **Next.js** and **FastAPI** on **PostgreSQL**, with development, staging and production environments defined in **Terraform**.

Software Engineer

Jul 2022 – Sep 2024

Quantitative Brokers

Chennai, India

- Designed a **real-time order monitoring system** processing **200 messages/sec** from multiple upstream sources into live aggregates and market data ladders for trading analysts.
- Cut risk control load time by **over 97%** (40 s to under 1 s) by moving data loading server-side and paginating.
- Reduced critical-path latency by **20%** and CPU usage by **50%** in a C# WPF order entry app consuming Bloomberg market data.
- Implemented **DV01** risk calculations for cash instruments, enabling the desk to expand its cash trading stream.
- Developed capture-and-replay tooling that reproduces production trading incidents from recorded message flow.

Software Development Intern

May 2021 – Jul 2021

Quantitative Brokers

Chennai, India

- Prototyped **OpenFin** interop across four trading applications and an Angular app plotting financial curves from a live **kdb+** database; both later shipped.

Android Developer

Feb 2021 – May 2021

Prescribe

Remote

- Developed a **Firestore**-backed Android app for patient intake and feedback, with a custom notification system and analytics on every step of the flow.

Software Development Intern

May 2020 – Jul 2020

Workfence Technologies

Remote

- Built a **geofencing attendance system** and a data visualisation interface designed to scale from 1,000 to **100,000 users**, plus a social-distancing Android app paired with distance-measuring hardware.

Projects

leak — Terminal Subscription Manager

github.com/RishikeshSreekumar/leak*Go, Bubble Tea, SQLite, GitHub Actions*

- Built a **local-first terminal app** that tracks and audits recurring subscriptions as a **single static Go binary**.
- Released versioned binaries for **six targets** (macOS, Linux and Windows on amd64 and arm64) through a Homebrew tap and install script, with CI on every push.

deadfall — Dead Component Detector

npmjs.com/package/deadfall*TypeScript, Node.js, ts-morph, npm*

- Published an **npm CLI** that maps a Next.js app's component usage graph and flags components nothing renders.
- Exits non-zero when dead components are found, so it doubles as a **CI gate**, and generates an interactive HTML report for exploring the import graph.

mailcleaner — Mailbox Cleanup Tool

github.com/RishikeshSreekumar/gmail_cleaner*Python, Textual, SQLite, IMAP, OAuth*

- Developed a **metadata-only** mailbox index across Gmail, Outlook, Fastmail, iCloud and any IMAP server, with terminal and browser UIs for bulk archive, trash and label actions.
- Designed a **no-permanent-delete** safety model: every bulk action previews a sample first, and archive and trash are refused on servers without IMAP MOVE rather than emulated.

Bansuri — Learn the Flute

flute.rishikeshs.dev*SvelteKit, TypeScript, Web Audio API*

- Created a flute learning app with **57 lessons** across 11 units and 8 practice tools, including **real-time pitch and drift detection** with all audio processed in the browser.
- Prerendered every lesson as a linkable, indexable URL, each opening its practice tool as a preset link.

Rubik's Cube Solver

React, Three.js, Web Workers, PWA

- Implemented **Kociemba's two-phase algorithm** in a Web Worker, with webcam sticker scanning and an animated, orbitable 3D solution that works offline as a PWA.
- Added cube-state validation that names why a cube is impossible: twisted corner, flipped edge or parity.

Technical Skills

Languages: TypeScript, JavaScript, Python, Go, C#, Java, SQL**Frontend:** React, Next.js, Angular, SvelteKit, Astro, Tailwind CSS, Three.js, WPF**Backend:** FastAPI, SQLAlchemy, Node.js, NestJS, .NET, REST APIs, WebSockets, Stripe**Applied AI:** LLMs, Retrieval-Augmented Generation (RAG), vector search, pgvector, Model Context Protocol (MCP)**Databases:** PostgreSQL, Redis, SQLite, kdb+**Cloud & DevOps:** Amazon Web Services (AWS), Microsoft Azure, Terraform, Docker, GitHub Actions, CI/CD, Vercel**Tools:** Git, Playwright, Alembic, Chrome Extensions, Web Workers, shell scripting

Education

Indian Institute of Technology Madras

2018 – 2022

Bachelor of Technology (B.Tech.), Chemical Engineering | CGPA: 8.58/10

Chennai, India

- **All India Rank 2,275** in JEE Advanced 2018; **State Rank 5** in KEAM 2018 (Kerala engineering entrance).